
STM32
CubeMX

1. Description

1.1. Project

Project Name	stm32-foc
Board Name	NUCLEO-F303RE
Generated with:	STM32CubeMX 6.18.1
Date	08/27/2026

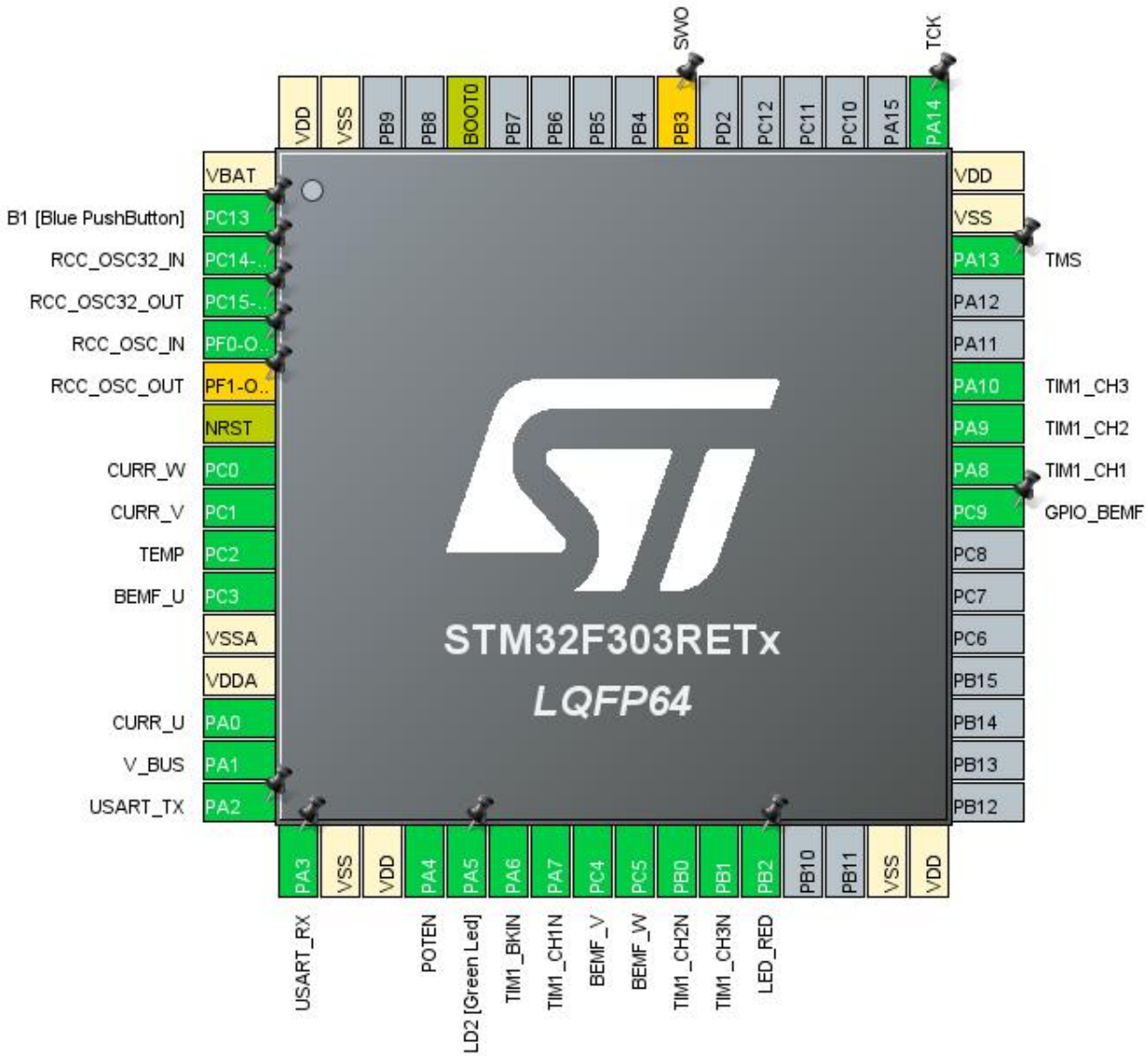
1.2. MCU

MCU Series	STM32F3
MCU Line	STM32F303
MCU name	STM32F303RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	Arm Cortex-M4

2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC13	I/O	GPIO_EXTI13	B1 [Blue PushButton]
3	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
4	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
5	PF0-OSC_IN	I/O	RCC_OSC_IN	
6	PF1-OSC_OUT *	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	PC0	I/O	ADC1_IN6	CURR_W
9	PC1	I/O	ADC1_IN7	CURR_V
10	PC2	I/O	ADC2_IN8	TEMP
11	PC3	I/O	ADC2_IN9	BEMF_U
12	VSSA	Power		
13	VDDA	Power		
14	PA0	I/O	ADC1_IN1	CURR_U
15	PA1	I/O	ADC1_IN2	V_BUS
16	PA2	I/O	USART2_TX	USART_TX
17	PA3	I/O	USART2_RX	USART_RX
18	VSS	Power		
19	VDD	Power		
20	PA4	I/O	ADC2_IN1	POTEN
21	PA5 **	I/O	GPIO_Output	LD2 [Green Led]
22	PA6	I/O	TIM1_BKIN	
23	PA7	I/O	TIM1_CH1N	
24	PC4	I/O	ADC2_IN5	BEMF_V
25	PC5	I/O	ADC2_IN11	BEMF_W
26	PB0	I/O	TIM1_CH2N	
27	PB1	I/O	TIM1_CH3N	
28	PB2 **	I/O	GPIO_Output	LED_RED
31	VSS	Power		
32	VDD	Power		
40	PC9 **	I/O	GPIO_Output	GPIO_BEMF
41	PA8	I/O	TIM1_CH1	
42	PA9	I/O	TIM1_CH2	
43	PA10	I/O	TIM1_CH3	
46	PA13	I/O	SYS_JTMS-SWDIO	TMS
47	VSS	Power		

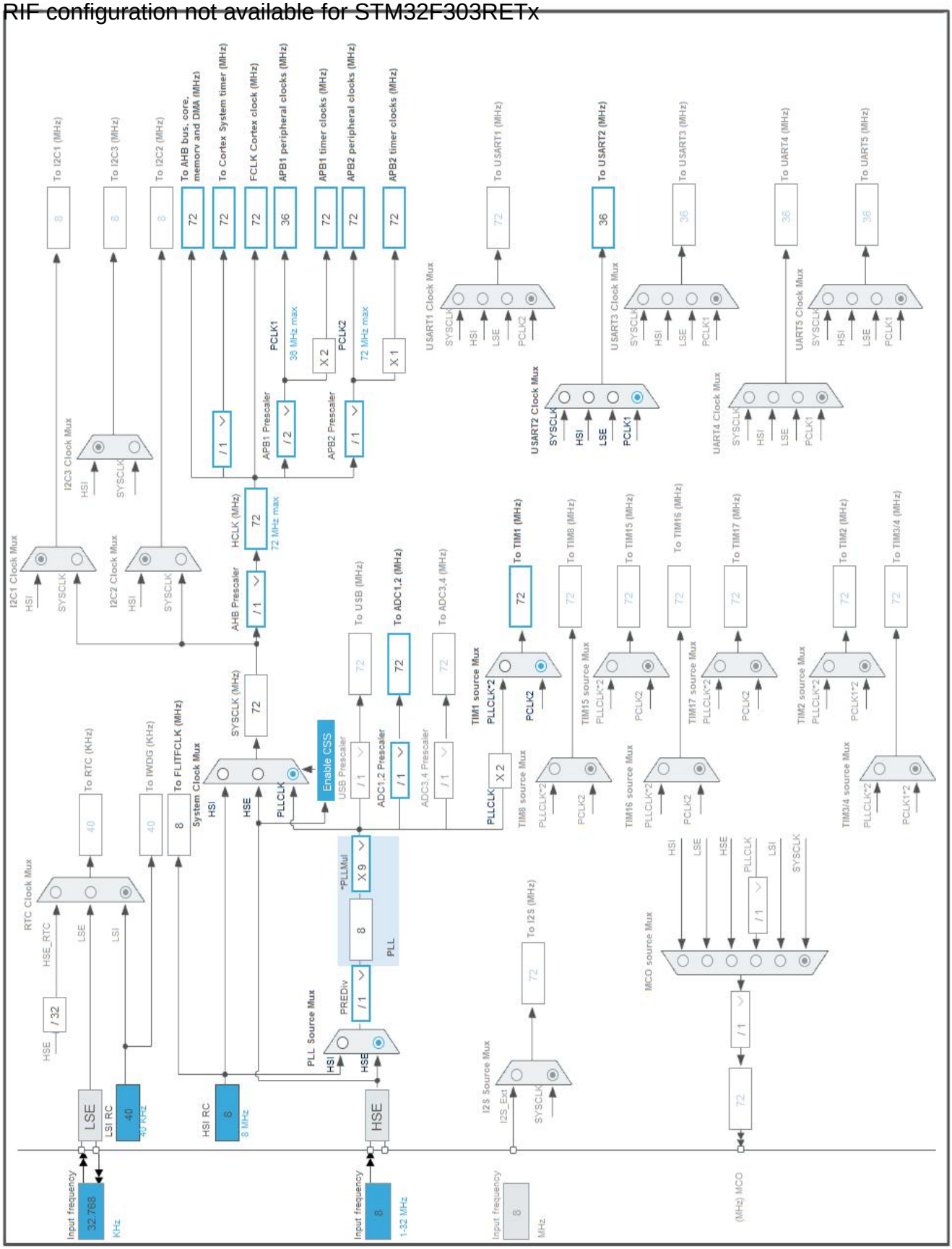
Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
48	VDD	Power		
49	PA14	I/O	SYS_JTCK-SWCLK	TCK
55	PB3 *	I/O	SYS_JTDO-TRACESWO	SWO
60	BOOT0	Boot		
63	VSS	Power		
64	VDD	Power		

** The pin is affected with an I/O function

* The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration

RIF configuration not available for STM32F303RETx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F3
Line	STM32F303
MCU	STM32F303RETx
Datasheet	DS10362_Rev5

1.2. Parameter Selection

Temperature	25
Vdd	3.6

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

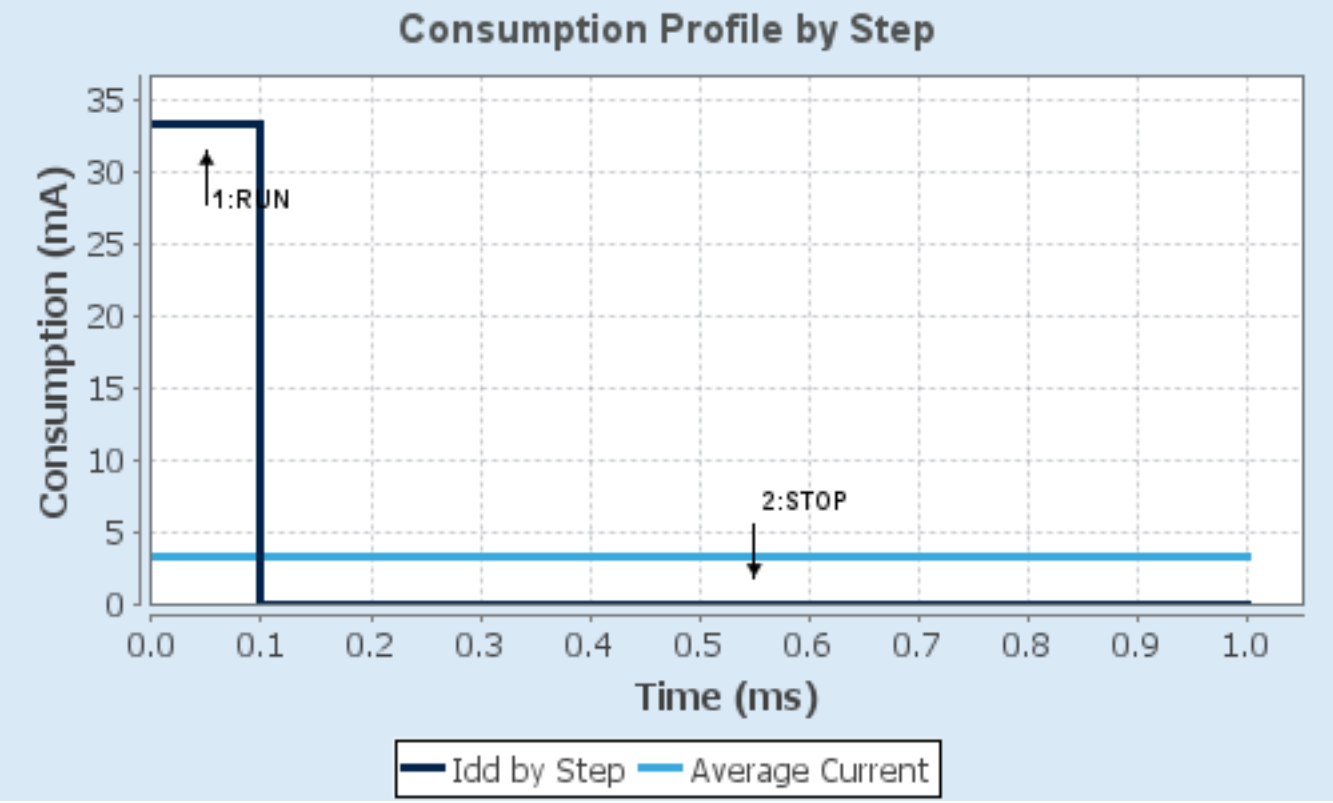
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.6	3.6
Voltage Source	Battery	Battery
Range	No Scale	No Scale
Fetch Type	FLASH	n/a
CPU Frequency	72 MHz	0 Hz
Clock Configuration	HSEBYP PLL	Regulator_LP
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	33.24 mA	9.8 µA
Duration	0.1 ms	0.9 ms
DMIPS	63.0	0.0
Ta Max	99.5	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	3.33 mA
Battery Life	1 month, 12 days, 1 hour	Average DMIPS	63.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
35
30
25
20
15
10
5
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	stm32-foc
Project Folder	C:\Users\akrap\stm32-foc
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F3 V1.11.6
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x600
Minimum Stack Size	0xC00

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_ADC1_Init	ADC1
4	MX_TIM1_Init	TIM1
5	MX_USART2_UART_Init	USART2
6	MX_ADC2_Init	ADC2

3. Peripherals and Middlewares Configuration

3.1. ADC1

IN1: IN1 Single-ended

IN2: IN2 Single-ended

IN6: IN6 Single-ended

IN7: IN7 Single-ended

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode

ADC_Settings:

Clock Prescaler	ADC Asynchronous clock mode
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Scan Conversion Mode	Enabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	**End of sequence of conversion ***
Overrun behaviour	Overrun data overwritten
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
SequencerNbRanks	1
Rank	1
Channel	**Channel 2 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset	0

ADC_Injected_ConversionMode:

Enable Injected Conversions	Enable
Number Of Conversions	**3 ***
External Trigger Source	**Timer 1 Capture Compare 4 event ***
External Trigger Conversion Edge	**Trigger detection on the falling edge ***
Injected Conversion Mode	None
Queue Injected Context Mode	Injected Queue enabled with Mode 0 (2 contexts, last active)

Rank	1
Channel	Channel 1
Sampling Time	1.5 Cycles
Offset Number	No offset
Injected Offset	0
Rank	**2 ***
Channel	**Channel 7 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Injected Offset	0
Rank	**3 ***
Channel	**Channel 6 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Injected Offset	0
Analog Watchdog 1:	
Enable Analog WatchDog1 Mode	false
Analog Watchdog 2:	
Enable Analog WatchDog2 Mode	false
Analog Watchdog 3:	
Enable Analog WatchDog3 Mode	false

3.2. ADC2

IN1: IN1 Single-ended

IN5: IN5 Single-ended

IN8: IN8 Single-ended

mode: IN9

IN11: IN11 Single-ended

3.2.1. Parameter Settings:

ADCs_Common_Settings:	
Mode	Independent mode
ADC_Settings:	
Clock Prescaler	ADC Asynchronous clock mode
Resolution	ADC 12-bit resolution
Data Alignment	Right alignment
Scan Conversion Mode	Enabled
Continuous Conversion Mode	Disabled

Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data overwritten
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Number Of Conversion	**2 ***
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
SequencerNbRanks	1
Rank	1
Channel	Channel 1
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset	0
Rank	**2 ***
Channel	**Channel 8 ***
Sampling Time	1.5 Cycles
Offset Number	No offset
Offset	0

ADC_Injected_ConversionMode:

Enable Injected Conversions	Enable
Number Of Conversions	0

Analog Watchdog 1:

Enable Analog WatchDog1 Mode	false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode	false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode	false

3.3. RCC

High Speed Clock (HSE): BYPASS Clock Source

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Prefetch Buffer	Enabled

Flash Latency(WS)	2 WS (3 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000

3.4. SYS

Debug: Serial Wire

Timebase Source: TIM17

3.5. TIM1

Clock Source : Internal Clock

Channel1: PWM Generation CH1 CH1N

Channel2: PWM Generation CH2 CH2N

Channel3: PWM Generation CH3 CH3N

Channel4: PWM Generation No Output

mode: Activate-Break-Input

3.5.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	**Center Aligned mode1 ***
Counter Period (AutoReload Register - 16 bits value)	**3600 ***
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	**Enable ***

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	**Update Event ***
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Enable
BRK Polarity	High
BRK Filter (4 bits value)	0

Break And Dead Time management - BRK2 Configuration:

BRK2 State	Disable
BRK2 Polarity	High

BRK2 Filter (4 bits value)	0

Break And Dead Time management - Output Configuration:

Automatic Output State	**Enable ***
Off State Selection for Run Mode (OSSR)	**Enable ***
Off State Selection for Idle Mode (OSSI)	**Enable ***
Lock Configuration	Off
Dead Time	**5 ***

Clear Input:

Clear Input Source	Disable

PWM Generation Channel 1 and 1N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	**Enable ***
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 2 and 2N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	**Enable ***
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 3 and 3N:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable
Fast Mode	**Enable ***
CH Polarity	High
CHN Polarity	High
CH Idle State	Reset
CHN Idle State	Reset

PWM Generation Channel 4:

Mode	PWM mode 1
Pulse (16 bits value)	0
Output compare preload	Enable

Fast Mode	**Enable ***
CH Polarity	High
CH Idle State	Reset

3.6. USART2

Mode: Asynchronous

3.6.1. Parameter Settings:

Basic Parameters:

Baud Rate	38400
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.7. FREERTOS

Interface: CMSIS_V2

3.7.1. Config parameters:

API:

FreeRTOS API	CMSIS v2

Versions:

FreeRTOS version	10.0.1
CMSIS-RTOS version	2.1.3

Kernel settings:

USE_PREEMPTION	Enabled
CPU_CLOCK_HZ	SystemCoreClock
TICK_RATE_HZ	1000
MAX_PRIORITIES	56
USE_SB_COMPLETED_CALLBACK	0
USE_MINI_LIST_ITEM	1
MINIMAL_STACK_SIZE	128
MAX_TASK_NAME_LEN	16
USE_16_BIT_TICKS	Disabled
IDLE_SHOULD_YIELD	Enabled
USE_MUTEXES	Enabled
USE_RECURSIVE_MUTEXES	Enabled
USE_COUNTING_SEMAPHORES	Enabled
QUEUE_REGISTRY_SIZE	8
USE_APPLICATION_TASK_TAG	Disabled
ENABLE_BACKWARD_COMPATIBILITY	Enabled
USE_PORT_OPTIMISED_TASK_SELECTION	Disabled
USE_TICKLESS_IDLE	Disabled
USE_TASK_NOTIFICATIONS	Enabled
RECORD_STACK_HIGH_ADDRESS	Disabled

Memory management settings:

Memory Allocation	Dynamic / Static
TOTAL_HEAP_SIZE	**32768 ***
Memory Management scheme	heap_4

Hook function related definitions:

USE_IDLE_HOOK	Disabled
USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Disabled
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Disabled

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	256

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

3.7.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled
xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Disabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled

3.7.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT	**Enabled ***

Project settings (see parameter description first):

Use FW pack heap file	Enabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PC0	ADC1_IN6	Analog mode	No pull-up and no pull-down	n/a	CURR_W
	PC1	ADC1_IN7	Analog mode	No pull-up and no pull-down	n/a	CURR_V
	PA0	ADC1_IN1	Analog mode	No pull-up and no pull-down	n/a	CURR_U
	PA1	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	V_BUS
ADC2	PC2	ADC2_IN8	Analog mode	No pull-up and no pull-down	n/a	TEMP
	PC3	ADC2_IN9	Analog mode	No pull-up and no pull-down	n/a	BEMF_U
	PA4	ADC2_IN1	Analog mode	No pull-up and no pull-down	n/a	POTEN
	PC4	ADC2_IN5	Analog mode	No pull-up and no pull-down	n/a	BEMF_V
	PC5	ADC2_IN11	Analog mode	No pull-up and no pull-down	n/a	BEMF_W
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PF0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	TMS
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	TCK
TIM1	PA6	TIM1_BKIN	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA7	TIM1_CH1N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB0	TIM1_CH2N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB1	TIM1_CH3N	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA8	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA9	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PA10	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USART2	PA2	USART2_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	USART_TX
	PA3	USART2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	USART_RX
Single Mapped Signals	PF1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
	PB3	SYS_JTDO-TRACESWO	n/a	n/a	n/a	SWO
GPIO	PC13	GPIO_EXTI13	**External Interrupt Mode with Falling edge trigger detection**	No pull-up and no pull-down	n/a	B1 [Blue PushButton]
	PA5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD2 [Green Led]
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED_RED

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPIO_BEMF

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
ADC1 and ADC2 interrupts	true	5	0
TIM1 trigger, commutation and TIM17 interrupts	true	15	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
TIM1 break and TIM15 interrupts	unused		
TIM1 update and TIM16 interrupts	unused		
TIM1 capture compare interrupt	unused		
USART2 global interrupt / USART2 wake-up interrupt through EXTI line 26	unused		
EXTI line[15:10] interrupts	unused		
Floating point unit interrupt	unused		

4.3.2. NVIC Code generation

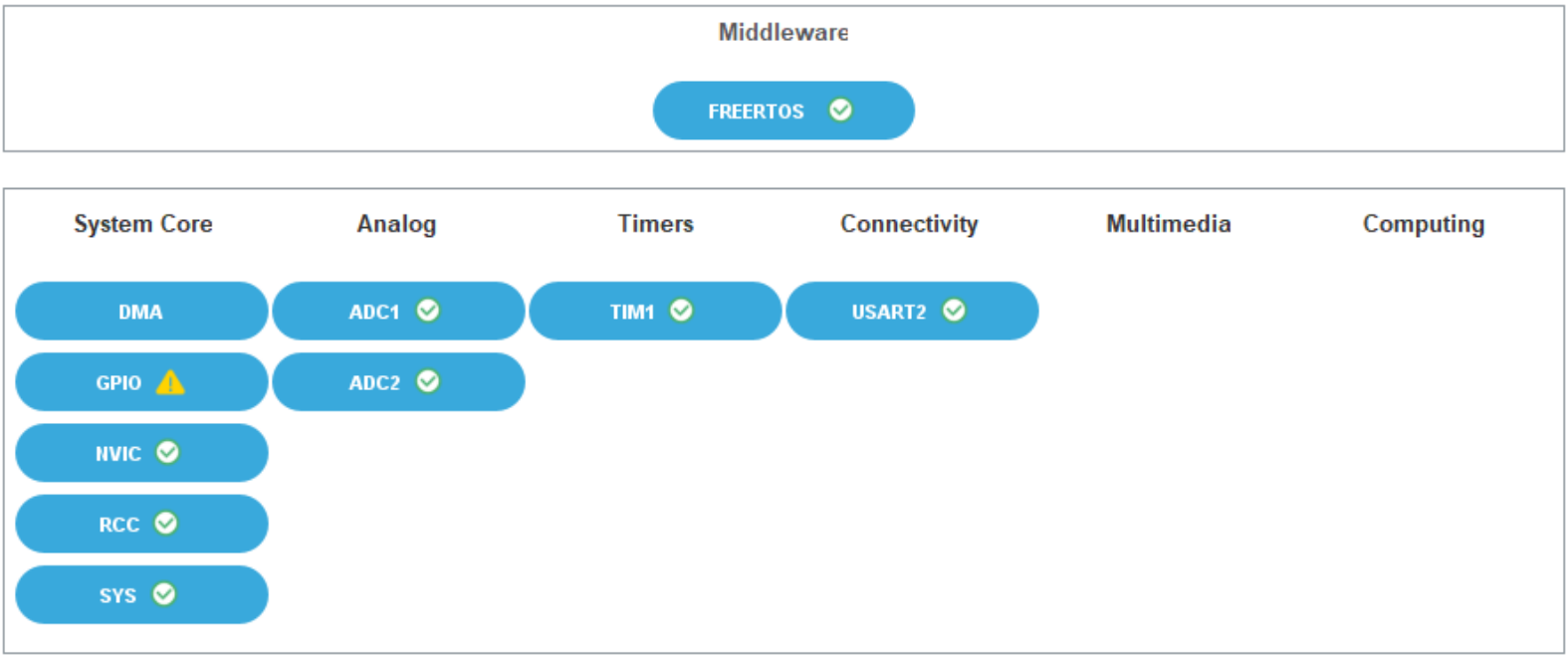
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
ADC1 and ADC2 interrupts	false	true	true
TIM1 trigger, commutation and TIM17 interrupts	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



Configuration stm32-foc
STM32CubeMX 6.18.1
Date 08/27/2026
MCU STM32F303RETx

PERIPHERALS	MODES	FUNCTIONS	PINS
ADC1	IN1 Single-ended	ADC1_IN1	PA0
ADC1	IN2 Single-ended	ADC1_IN2	PA1
ADC1	IN6 Single-ended	ADC1_IN6	PC0
ADC1	IN7 Single-ended	ADC1_IN7	PC1
ADC2	IN1 Single-ended	ADC2_IN1	PA4
ADC2	IN5 Single-ended	ADC2_IN5	PC4
ADC2	IN8 Single-ended	ADC2_IN8	PC2
ADC2	IN9 Single-ended	ADC2_IN9	PC3
ADC2	IN11 Single-ended	ADC2_IN11	PC5
RCC	BYPASS Clock Source	RCC_OSC_IN	PF0-OSC_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_IN	PC14-OSC32_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_OUT	PC15-OSC32_OUT
SYS	Serial Wire	SYS_JTCK-SWCLK	PA14
SYS	Serial Wire	SYS_JTMS-SWDIO	PA13
SYS	TIM17	SYS_VS_tim17	VP_SYS_VS_tim17
TIM1	Internal Clock	TIM1_VS_ClockSourceINT	VP_TIM1_VS_ClockSourceINT
TIM1	PWM Generation CH1 CH1N	TIM1_CH1	PA8
TIM1	PWM Generation CH1 CH1N	TIM1_CH1N	PA7
TIM1	PWM Generation CH2 CH2N	TIM1_CH2	PA9
TIM1	PWM Generation CH2 CH2N	TIM1_CH2N	PB0
TIM1	PWM Generation CH3 CH3N	TIM1_CH3	PA10
TIM1	PWM Generation CH3 CH3N	TIM1_CH3N	PB1
TIM1	PWM Generation No Output	TIM1_VS_no_output4	VP_TIM1_VS_no_output4
TIM1	Activate-Break-Input	TIM1_BKIN	PA6
USART2	Asynchronous	USART2_RX	PA3
USART2	Asynchronous	USART2_TX	PA2

Pin_Nb	PINs	FUNCTIONs	LABELs
2	PC13	GPIO_EXTI13	B1 [Blue PushButton]
3	PC14-OSC32_IN	RCC_OSC32_IN	
4	PC15-OSC32_OUT	RCC_OSC32_OUT	
5	PF0-OSC_IN	RCC_OSC_IN	
6	PF1-OSC_OUT*	RCC_OSC_OUT	
8	PC0	ADC1_IN6	CURR_W
9	PC1	ADC1_IN7	CURR_V
10	PC2	ADC2_IN8	TEMP
11	PC3	ADC2_IN9	BEMF_U
14	PA0	ADC1_IN1	CURR_U
15	PA1	ADC1_IN2	V_BUS
16	PA2	USART2_TX	USART_TX
17	PA3	USART2_RX	USART_RX
20	PA4	ADC2_IN1	POTEN
21	PA5	GPIO_Output	LD2 [Green Led]
22	PA6	TIM1_BKIN	
23	PA7	TIM1_CH1N	
24	PC4	ADC2_IN5	BEMF_V

25	PC5	ADC2_IN11	BEMF_W
26	PB0	TIM1_CH2N	
27	PB1	TIM1_CH3N	
28	PB2	GPIO_Output	LED_RED
40	PC9	GPIO_Output	GPIO_BEMF
41	PA8	TIM1_CH1	
42	PA9	TIM1_CH2	
43	PA10	TIM1_CH3	
46	PA13	SYS_JTMS-SWDIO	TMS
49	PA14	SYS_JTCK-SWCLK	TCK
55	PB3*	SYS_JTDO-TRACESWO	SWO

PERIPHERALS	MODES	FUNCTIONS	PINS
ADC1	IN1 Single-ended	ADC1_IN1	PA0
ADC1	IN2 Single-ended	ADC1_IN2	PA1
ADC1	IN6 Single-ended	ADC1_IN6	PC0
ADC1	IN7 Single-ended	ADC1_IN7	PC1
ADC2	IN1 Single-ended	ADC2_IN1	PA4
ADC2	IN5 Single-ended	ADC2_IN5	PC4
ADC2	IN8 Single-ended	ADC2_IN8	PC2
ADC2	IN9 Single-ended	ADC2_IN9	PC3
ADC2	IN11 Single-ended	ADC2_IN11	PC5
RCC	BYPASS Clock Source	RCC_OSC_IN	PF0-OSC_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_IN	PC14-OSC32_IN
RCC	Crystal/Ceramic Resonator	RCC_OSC32_OUT	PC15-OSC32_OUT
SYS	Serial Wire	SYS_JTCK-SWCLK	PA14
SYS	Serial Wire	SYS_JTMS-SWDIO	PA13
SYS	TIM17	SYS_VS_tim17	VP_SYS_VS_tim17
TIM1	Internal Clock	TIM1_VS_ClockSourceINT	VP_TIM1_VS_ClockSourceINT
TIM1	PWM Generation CH1 CH1N	TIM1_CH1	PA8
TIM1	PWM Generation CH1 CH1N	TIM1_CH1N	PA7
TIM1	PWM Generation CH2 CH2N	TIM1_CH2	PA9
TIM1	PWM Generation CH2 CH2N	TIM1_CH2N	PB0
TIM1	PWM Generation CH3 CH3N	TIM1_CH3	PA10
TIM1	PWM Generation CH3 CH3N	TIM1_CH3N	PB1
TIM1	PWM Generation No Output	TIM1_VS_no_output4	VP_TIM1_VS_no_output4
TIM1	Activate-Break-Input	TIM1_BKIN	PA6
USART2	Asynchronous	USART2_RX	PA3
USART2	Asynchronous	USART2_TX	PA2

Pin Nb	PINs	FUNCTIONs	LABELs
2	PC13	GPIO_EXTI13	B1 [Blue PushButton]
3	PC14-OSC32_IN	RCC_OSC32_IN	
4	PC15-OSC32_OUT	RCC_OSC32_OUT	
5	PF0-OSC_IN	RCC_OSC_IN	
6	PF1-OSC_OUT*	RCC_OSC_OUT	
8	PC0	ADC1_IN6	CURR_W
9	PC1	ADC1_IN7	CURR_V
10	PC2	ADC2_IN8	TEMP
11	PC3	ADC2_IN9	BEMF_U
14	PA0	ADC1_IN1	CURR_U

15	PA1	ADC1_IN2	V_BUS
16	PA2	USART2_TX	USART_TX
17	PA3	USART2_RX	USART_RX
20	PA4	ADC2_IN1	POTEN
21	PA5	GPIO_Output	LD2 [Green Led]
22	PA6	TIM1_BKIN	
23	PA7	TIM1_CH1N	
24	PC4	ADC2_IN5	BEMF_V
25	PC5	ADC2_IN11	BEMF_W
26	PB0	TIM1_CH2N	
27	PB1	TIM1_CH3N	
28	PB2	GPIO_Output	LED_RED
40	PC9	GPIO_Output	GPIO_BEMF
41	PA8	TIM1_CH1	
42	PA9	TIM1_CH2	
43	PA10	TIM1_CH3	
46	PA13	SYS_JTMS-SWDIO	TMS
49	PA14	SYS_JTCK-SWCLK	TCK
55	PB3*	SYS_JTDO-TRACESWO	SWO